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Dated : May 20, 2015

Mr. Narayan Bhardwaj,

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Dear Mr. Bhardwaj,

This is with reference to the testing performed on May 16, 2015 for micro turbines (i) VARUN-I and (ii) VARUN-III developed by MACLEC TECHNICAL PROJECT LABORATORY PVT. LTD. These turbines were installed in the water stream at tail race channel of Ramganga Dam at Kalagarh. Based on the observations and measurements made on the site, it is found that the system was nicely installed in the stream and the performance of the system can be further improved provided some more modification can be made in these turbines. The data measured during the testing are as given below;

1. VARUN-I (Undershot water wheel type Hydrokinetic Turbine).

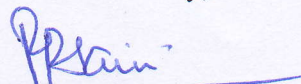
Water velocity at inlet of turbine (m/s)	Cross section area at inlet (m ²)	Turbine RPM	Load (W)
1.2	0.525	30	126

2. VARUN-III (Conical spiral shaped hydrokinetic turbine).

Water velocity at inlet of turbine (m/s)	Cross section area at inlet (m ²)	Turbine RPM	Load (W)
1.3	0.6	28	132

Thanking you,

Yours faithfully,


(R.P. Saini)